What Makes a Successful Video Game?

A Data-Driven Analysis of Player Reception and Commercial Performance

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Basic Information

- **Course:** COMP-248 – Project in Data Science
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- **Academic Year:** Year 2, BSc in Data Science

1. Background and Motivation

The video game industry is one of the largest and fastest-growing entertainment industries worldwide, generating billions of dollars in revenue every year. Despite this growth, not all video games achieve the same level of success. While some titles become global hits with strong sales and positive player reception, others fail to attract attention even when backed by large studios.

As a data science student with an interest in gaming, I am motivated to explore what factors contribute to a video game's success. Understanding these factors can be valuable to developers, publishers, and players alike. From a data science perspective, the gaming industry offers rich, real-world datasets suitable for exploratory data analysis, visualization, and basic machine learning.

2. Project Objectives (Hypotheses)

The main objective of this project is to analyze historical video game data to identify patterns associated with commercial and player success.

The project aims to answer the following questions:

1. Which video game attributes (such as genre, platform, and release year) are most strongly associated with high global sales?
2. Is there a relationship between user and critic ratings and commercial success?
3. Can a simple machine learning model classify games as successful or unsuccessful based on a small set of features?

Hypothesis:
Video games with higher user ratings and belonging to specific genres (e.g., Action or RPG) are more likely to achieve higher global sales. Additionally, simple classification models can reasonably distinguish successful games from unsuccessful ones.

3. Data Sources

This project will integrate multiple recent, publicly available real-world datasets related to the video game industry.

Primary Data Source

- **Video Game Sales 2024 Dataset (Kaggle)**
  Link: https://www.kaggle.com/datasets/asaniczka/video-game-sales-2024

  This dataset contains global sales information for tens of thousands of video games, including attributes such as genre, platform, publisher, release year, and regional/global sales figures. It will serve as the core dataset for analyzing commercial success.

Secondary Data Sources

- **Metacritic Games and Reviews Dataset (2025)**
  Link: https://www.kaggle.com/datasets/davutb/metacritic-games

  This dataset provides critic and user review scores for video games, including more recent releases. It will be used to examine the relationship between player and critic reception and sales performance.

- **Steam Games Dataset (2025)**
  Link: https://www.kaggle.com/datasets/artermiloff/steam-games-dataset

  This dataset includes metadata for modern PC games available on Steam, such as tags, categories, release dates, pricing, and aggregated user review information. It will be used to enrich genre analysis and study player engagement trends.

All datasets are provided in CSV format and may require integration using common attributes such as game title and release year. The use of multiple datasets is intended to enhance exploratory analysis.

4. Data Modeling

Each row in the dataset represents a single video game.

Key attributes include:

- **Categorical variables:** Genre, Platform, Publisher
- **Numerical variables:** Global sales, release year, rating scores

A derived target variable will be created:

- **Success Label:** A binary label (e.g., High vs Low success), defined using a sales threshold such as the median or a percentile-based cutoff.

5. Data Preprocessing

The following preprocessing steps are expected: • Handling missing values in ratings and sales data • Removing duplicates and irrelevant records • Encoding categorical variables (e.g., genre and platform) • Scaling numerical features if required Preprocessing will ensure the dataset is clean, structured, and ready for analysis.

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6. Data Analysis Plan

6.1 Exploratory Data Analysis (EDA)

The EDA phase will focus on understanding the dataset and identifying trends. Planned analyses include:

- Distribution of video game sales across genres and platforms
- Sales trends over time
- Relationship between ratings and sales
- Identification of top-performing genres and platforms

6.2 Machine Learning (Optional / Lightweight)

To extend the analysis, simple machine learning models will be applied:

- **Models:** Logistic Regression, Decision Tree
- **Task:** Binary classification (successful vs unsuccessful games)
- **Evaluation Metrics:** Accuracy and confusion matrix

7. Data Visualization

To enhance data storytelling and produce professional, interactive visuals, this project will use **Power BI** as the primary visualization tool.

Planned visualizations include:

- Interactive dashboards showing global sales distribution by genre, platform, and year
- Line charts illustrating sales trends over time
- Bar charts comparing average sales across genres and platforms
- Slicers and filters to allow dynamic exploration of the data

Power BI will be used after data preprocessing and basic analysis in Python. Cleaned datasets will be exported and loaded into Power BI to create dashboards that clearly communicate insights to both technical and non-technical audiences.

8. Tools and Libraries

The following tools and libraries will be used:

- **Python**
- **Pandas & NumPy** – data handling and preprocessing
- **Scikit-learn** – machine learning models
- **Power BI Desktop** – data visualization
- **Jupyter Notebook** – project documentation and presentation

9. Expected Outcomes

Expected outcomes include:

- Working with real-world gaming datasets
- Strong exploratory data analysis

- Simple and explainable machine learning models
- Clear and professional visual storytelling using Power BI